

Annual Climate Change Impact Report 2025

Executive Summary

Global average surface temperatures in 2024 exceeded pre-industrial baselines by 1.45 °C, placing the world perilously close to the 1.5 °C threshold established by the Paris Agreement. This report synthesises observational data, climate model projections, and socioeconomic impact assessments to inform policy decisions for the coming decade.

Key findings indicate accelerating ice-sheet loss in Greenland and West Antarctica, record ocean heat content through the full water column, and a statistically significant increase in the frequency and intensity of extreme precipitation events across all inhabited continents.

1. Atmospheric CO₂ Concentrations

Monthly mean CO₂ concentration at Mauna Loa Observatory reached 424.3 ppm in May 2024, representing a year-on-year increase of 2.8 ppm. The decadal growth rate has accelerated from 1.5 ppm/year in the 1990s to 2.5 ppm/year in the 2020s, driven by continued fossil-fuel combustion and land-use change, partially offset by natural carbon sinks.

Permafrost thaw in Arctic regions is emerging as a significant positive feedback loop. Decomposition of organic matter previously locked in frozen soils releases methane (CH₄), a greenhouse gas with a 20-year global warming potential 86 times that of CO₂. Current Earth system models may underestimate permafrost emissions by up to 40%.

2. Sea Level Rise and Coastal Impacts

Global mean sea level rose 3.7 mm in 2024, bringing the cumulative rise since 1993 to approximately 112 mm. Thermal expansion of warming ocean water accounts for roughly 40% of this increase, while the remainder stems from land-ice melt. Under RCP 8.5 scenarios, sea levels could rise between 0.61 m and 1.10 m by 2100.

Low-lying coastal megacities — including Miami, Jakarta, Mumbai, and Shanghai — face compounding risks from sea-level rise, storm surge intensification, and saltwater intrusion into freshwater aquifers. The economic cost of unmitigated coastal flooding is projected to reach USD 14.2 trillion annually by 2100 under a business-as-usual emissions trajectory.

Managed retreat strategies, nature-based defences such as mangrove restoration, and engineered solutions including seawalls and surge barriers must be deployed in complementary combinations tailored to local geomorphology, population density, and financial capacity.

3. Extreme Weather Events

The attribution science linking anthropogenic climate change to specific extreme weather events has matured substantially. In 2024, 23 of 28 peer-reviewed rapid attribution studies found that human-caused warming made the studied event more likely or more intense, with median probability ratios exceeding 2.0 for heatwaves and heavy precipitation episodes.

Tropical cyclone intensification rates have increased. The proportion of storms reaching Category 4–5 intensity has grown by approximately 25% since the 1980s. Rapid intensification events — defined as a wind speed increase of at least 35 knots in 24 hours — have become more frequent in the Atlantic and Western Pacific basins, complicating evacuation planning.

4. Mitigation and Adaptation Strategies

Achieving net-zero emissions by 2050 requires an immediate tripling of renewable energy capacity additions, electrification of heating and transport, and deployment of carbon dioxide removal (CDR) technologies at gigaton scale. Bioenergy with carbon capture and storage (BECCS) and direct air capture (DAC) are the leading CDR approaches, though both face significant land-use, energy, and cost constraints at the scales required.

Adaptation investment remains critically underfunded. The UN Environment Programme estimates that developing nations require USD 387 billion annually for adaptation by 2030, yet current international climate finance flows for adaptation total only USD 63 billion per year. Closing this gap is both a moral imperative and an economic necessity, as every dollar invested in resilient infrastructure returns an estimated USD 6 in avoided disaster losses.